Muhammad Hassin Saghir

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Summary

Undergraduate Computer Systems Engineer with hands-on internship experience in AI engineering and full-stack development. Experienced in building LLM-powered agentic systems, RAG assistants, machine learning models, and web applications with REST APIs, automated testing, and CI/CD, with a focus on clean, scalable, maintainable code.

Experience

NCBC Karachi, Pakistan

AI Engineer Intern *Jun 2026 – Sept 2026*

- Developed an LLM-powered agentic system that interprets natural language queries and autonomously selects and executes the right tools to deliver data-driven insights
- Enabled natural language interaction with complex multi-layer datasets, translating user intent into structured analysis and visualization requests
- Designed multi-step agent workflows with intelligent dataset orchestration, enabling reasoning over historical and time-aware data
- Integrated LLM agents with a modular, provider-based backend to support scalable, extensible AI-driven analysis

10Pearls Karachi, Pakistan

Software Engineer Intern *Sept 2025 – Dec 2025*

- Engineered full-stack web applications with responsive user interfaces and scalable REST APIs
- Applied modern state management, error handling, and production-ready development practices to deliver maintainable, reliable features
- Wrote unit and component tests to improve application reliability
- Established code quality analysis and CI/CD pipelines to streamline builds and deployments

Projects

TerraFlux – Agentic Geospatial Weather Visualization System GitHub

LangGraph, LangChain, GeoPandas, Shapely

- Built an agent-driven workflow that converts natural-language weather questions into structured geospatial requests, connecting location resolution, weather-data processing, and interactive map generation
- Implemented hierarchical administrative-boundary resolution from country to district, with ordered source fallbacks and escalation to a broader parent region when a precise boundary cannot be resolved
- Developed a climate analysis workflow that processes historical weather data into regional statistics and interactive, layer-based maps, with support for regional comparisons and forecast retrieval

GitSense – Autonomous GitHub Code Review GitHub

Python, FastAPI, LangGraph, Redis, PostgreSQL, Langfuse, Docker

- Automated code reviews by checking every GitHub commit and pull request as it happens, flagging security risks, breaking changes, and bugs without manual effort
- Designed an automated fix generator that explains detected problems in plain English and provides ready-to-copy code solutions, cutting down manual debugging time
- Ran several specialized checks in parallel, covering security, reliability, code quality, and breaking changes, then combined them into one overall risk score while filtering out unreliable AI-generated findings

Bitcoin Market Direction Prediction GitHub

LightGBM, LSTM, Optuna, scikit-learn

- Trained a model to predict Bitcoin's next market move and tested it on unseen future data, earning an average +30.1% return and cutting the worst loss from 52.2% to 17.8% versus buy-and-hold
- Optimized results by automatically tuning model settings and adjusting trade signal thresholds, turning a profit in 4 of 5 test periods with a peak return of +95.8%
- Discovered that how outcomes were labeled mattered more than model choice, and adjusting that threshold cut prediction error nearly in half

Agentic RAG – Military Doctrine Q&A Assistant GitHub

Python, FastMCP, LangChain, LangGraph, ImageKit

- Built a chat assistant that answers tactical questions by searching military doctrine manuals with an AI agent and citing the exact source
- Designed the agent's multi-step reasoning and document indexing so new manuals load in the background without one failure stopping the system
- Combined keyword and meaning-based search to improve retrieval accuracy, and added a review step that checks retrieved passages before generating an answer

TeamSync – Real-Time Team Chat Workspace GitHub

Node.js, Express.js, MongoDB, React, Socket.IO, ImageKit, JWT, Zustand

- Constructed a real-time team collaboration platform with separate workspaces, role-based access, live chat with typing indicators, online status and threads, and secure login with email OTP verification
- Introduced a file management system with folder hierarchy, multi-user assignment, and cloud-based file delivery, alongside an admin dashboard with analytics for user, workspace, and content moderation
- Structured the backend with clear separation between authentication, workspace, messaging, and file logic, and documented REST endpoints for easy integration

ClassConnect – Digital Classroom Workflow Tool GitHub

Python, Django, Tailwind CSS, MySQL

- Contributed to a virtual classroom platform where teachers create classrooms and post assignments and announcements, and students join classes to view updates and stay organized
- Helped deliver assignment submissions, comment discussions on posts and student work, and secure sign-up and login for both teachers and students
- Kept the backend and database structure organized and easy to extend as new classroom features are added

SalaryScout – Multi-Source Job Discovery System GitHub

JavaScript, React, Vite, Tailwind CSS, shadcn/ui, Radix UI, Axios, React Router, Framer Motion

- Assembled a full-stack job finder application integrating multiple external APIs for real-time job listings and salary estimation, with dynamic filtering by experience level, location type, and company
- Crafted a single, easy-to-browse job feed that brings listings together in one place and gives every posting a direct application link, so users can apply in one step
- Refined a clean, mobile-friendly layout so users can search and compare jobs comfortably on any device

Education

NED University Of Engineering and Technology Karachi, Pakistan

B.E Computer Systems Engineering *Oct. 2023 – Ongoing*

CGPA: 3.512

Certifications

Agentic AI with LangGraph, CrewAI, AutoGen and BeeAI – IBM Certificate

Agentic AI with LangChain and LangGraph – IBM Certificate

Technical Skills

Languages: Python, SQL, NoSQL (MongoDB), JavaScript, TypeScript
Frameworks: React, Node.js, Express, NextJS, FastAPI, Socket.IO, ShadCN UI, Framer Motion
Machine Learning: PyTorch, TensorFlow, scikit-learn, LightGBM, Optuna — CNNs, RNNs, LSTMs
AI: LangChain, LangGraph, RAG, Agentic AI, Model Context Protocol (MCP), LLM Integration
Libraries: pandas, NumPy, Matplotlib, GeoPandas, Folium, Tanstack Query, Zustand
Developer Tools: Git, GitHub Actions, Docker, Linux (Ubuntu), SonarQube, Jest, React Testing Library, REST APIs